

CMP-17 - Trust Service

Produce multidimensional, evidence-based, scoped, expiring Trust Assessments without granting authority or permission.

Authority and state ownership

Book I: Decisions 24-29, 35, 40-43, 47-50, and 58. Book II: Chapters 18. Authoritative state: Trust Assessments, trust dimensions, confidence and expiry metadata, trust-policy inputs. Explicitly non-owned: Evidence Objects, Authority, Permission, Treaties, authoritative Identity state.

Critical invariant

Trust may inform policy but never grants Authority, Permission, identity, or Treaty scope.

Logical interfaces

CMP-17-IF-01 / Command / Request Trust Assessment

Command `Request Trust Assessment` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-17-IF-02 / Command / Update Trust Dimension

Command `Update Trust Dimension` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-17-IF-03 / Command / Invalidate Assessment

Command `Invalidate Assessment` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-17-IF-04 / Query / Get Trust Assessment

Query `Get Trust Assessment` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-17-IF-05 / Query / Explain Trust Basis

Query `Explain Trust Basis` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-17-IF-06 / Query / List Expiring Assessments

Query `List Expiring Assessments` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-17-IF-07 / Event / Trust Assessed

Event `Trust Assessed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-17-IF-08 / Event / Trust Degraded

Event `Trust Degraded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-17-IF-09 / Event / Assessment Expired

Event `Assessment Expired` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Requested > Assessed > Degraded > Expired or Revoked. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: trust-as-authority, stale evidence, dimension collapse, reputation contagion, expiry bypass. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-17-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-17-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-17-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-17-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-17-TST-005 - Critical invariant. Trust may inform policy but never grants Authority, Permission, identity, or Treaty scope. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-17-TST-006 - Failure containment. Inject trust-as-authority, stale evidence, dimension collapse, reputation contagion, expiry bypass. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-17-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-17-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-17-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-17-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.